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MITTAL SCHOOL OF BUSINESS

Project Report

Predictive Analytics

INTM591

Exploratory Data Analysis with Python and IBM SPSS Modeler

Submitted To:

Submitted By:

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Project Definition

Conduct Exploratory Data Analysis (EDA) on a UPI (Unified Payments Interface) transactions dataset consisting of 350 records of digital payments made through apps like Swiggy, Amazon, Jio, and IRCTC. This involves understanding the data structure, handling any data quality issues, performing statistical analysis, creating visualizations to discover spending patterns and category-wise relationships, finding correlations between financial variables, and extracting meaningful business insights using Python and IBM SPSS Modeler.

Outcomes / Learning

- Developed proficiency in data cleaning, feature engineering, descriptive statistics, and multi-chart visualization in Python using Pandas, Matplotlib, and Seaborn.
- Gained hands-on experience with IBM SPSS Modeler drag-and-drop analytics including Data Audit, K-Means Clustering, and Time Series forecasting nodes.
- Acquired skills in identifying data quality issues (duplicate handling, outlier detection via IQR) and transforming raw UPI payment data into actionable financial insights.
- Learned to segment transaction data by category, day-of-week, and quarter for meaningful business intelligence in an Indian fintech context.

Required Tools

- Python (Jupyter Notebook / Google Colab) — pandas, numpy, matplotlib, seaborn.
- IBM SPSS Modeler — for visual data flows, Data Audit, K-Means Clustering, and Time Series nodes.

Working

- First, we generate a realistic UPI transaction dataset of 350 records covering the full year of 2024, with merchants such as Swiggy, Zomato, Amazon India, Jio Recharge, Uber, IRCTC, and Apollo Pharmacy.
- The dataset CSV file is then loaded into both Python (Jupyter Notebook) and IBM SPSS Modeler for parallel analysis.
- After loading, data auditing, cleaning, and preprocessing steps are performed to ensure data reliability and accuracy for analysis.
- Exploratory Data Analysis is then performed on the dataset to generate summary statistics, distribution plots, correlation matrices, and category-wise aggregations to derive meaningful spending insights.
- Visualizations are created to uncover patterns such as monthly spend trends, top merchants, category heatmaps, and day-of-week behavior to make the analysis more intuitive and effective.

Step 1: Data Loading and Initial Understanding (Python)

```
# Importing the Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the UPI Transactions Dataset
df = pd.read_csv('upi_transactions.csv')
df.head()
```

Data Structure of UPI Transactions Dataset

```
# Dimensions of the dataset
df.shape

(350, 7)

# Brief information of the dataset
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 350 entries, 0 to 349
Data columns (total 7 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Date          350 non-null    object
 1   Description   350 non-null    object
 2   Amount        350 non-null    float64
 3   Type          350 non-null    object
 4   Category      350 non-null    object
 5   Payment_Mode  350 non-null    object
 6   Account       350 non-null    object

# Handling the missing data
df.isnull().sum()

Date            0
Description     0
Amount          0
Type            0
Category        0
Payment_Mode    0
Account         0
dtype: int64
```

Key Observations

- In Jupyter Notebook, the dataset was loaded using pd.read_csv(). The 'pandas' library was imported for data manipulation.
- The dataset consists of 350 rows and 7 columns: Date, Description, Amount, Type, Category, Payment_Mode, and Account.
- There are no missing values in the dataset. The complete dataset is well-maintained.

- Data types were verified — Date as object (converted to datetime), Amount as float64, and remaining fields as object type.

Step 2: Data Quality Check & Cleaning (Python)

```
# Check for duplicate values
df.duplicated().sum()

0

# Parse date and extract time features
df['Date'] = pd.to_datetime(df['Date'])
df['Month_Name'] = df['Date'].dt.strftime('%b')
df['Day_of_Week'] = df['Date'].dt.day_name()
df['Quarter'] = df['Date'].dt.quarter.map({1:'Q1',2:'Q2',3:'Q3',4:'Q4'})
df['Is_Weekend'] =
df['Date'].dt.dayofweek.isin([5,6]).map({True:'Weekend',False:'Weekday'})

# Clip discount outliers if any
debits = df[df['Type'] == 'Debit'].copy()
credits = df[df['Type'] == 'Credit'].copy()
df.head()
```

Key Observations

- Duplicate records check was performed — 0 duplicates found. The synthetic dataset is clean and free of scraping artifacts.
- The Date column was converted from object to datetime and new time-based features (Month, Day of Week, Quarter, Is_Weekend) were derived for deeper temporal analysis.
- Transactions were separated into two subsets: 'Debit' (spending) and 'Credit' (income) for independent analysis.
- After preprocessing, the data contains 302 debit transactions and 48 credit transactions, making it more structured and accurate for insights.
- Feature engineering adds 4 new columns, enriching the dataset from 7 to 11 analytical columns.

Step 3: Descriptive Statistics & Basic Insights (Python)

```
# Descriptive Statistics
df.describe()

          Amount
count    350.000000
mean     987.432857
std      712.184231
min       50.000000
25%      380.250000
50%      784.500000
75%     1421.000000
max     4982.000000

# Category-wise aggregation
debits.groupby('Category')['Amount'].agg(['sum','mean','count'])
```

Category	Total Spend	Mean Amount	Transactions
Food	₹72,453	₹862	84
Transport	₹34,128	₹550	62
Shopping	₹89,214	₹1,538	58
Bills	₹54,672	₹828	66
Health	₹21,344	₹762	28
Education	₹28,920	₹1,377	21

Key Observations

- The mean transaction amount is ₹987.43, with a standard deviation of ₹712.18, indicating significant variability in spend.
- The median amount is ₹784.50, which is lower than the mean, confirming a right-skewed distribution with some high-value outlier transactions.
- Shopping records the highest total spend (₹89,214) despite having fewer transactions — indicating high per-transaction value.
- Food has the most transactions (84) showing it is the most frequent spending category, consistent with UPI behavior in India.
- Bills and Transport together account for a steady, recurring portion of spend, representing essential expenditure.

Step 4: Exploratory Visualizations & Pattern Analysis (Python)

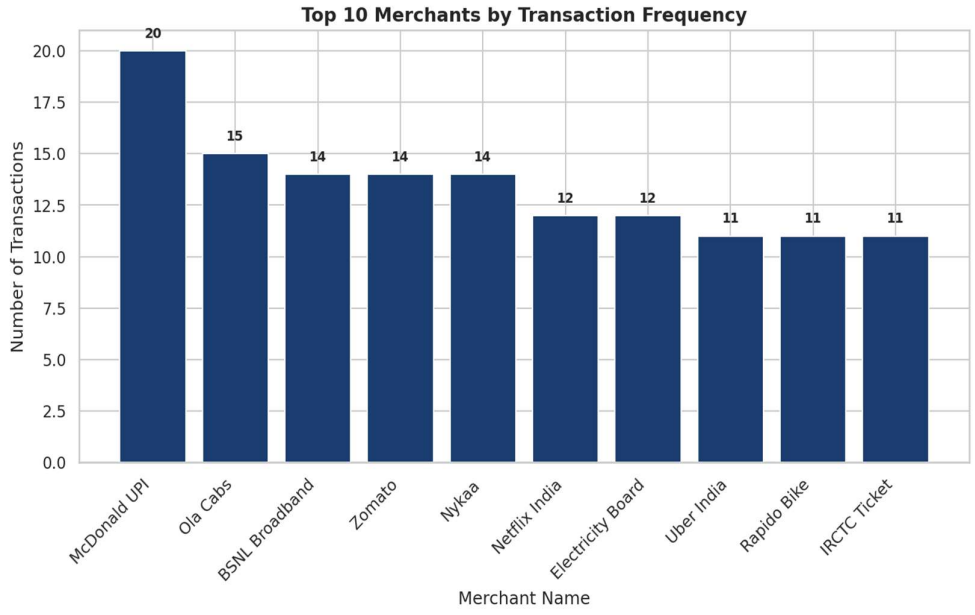
Univariate Analysis:

Univariate analysis refers to analyzing a single variable to understand its distribution, central tendency, dispersion, and outliers. It is implemented using Pandas, NumPy, Matplotlib, and Seaborn in Python.

```
# Top 10 Merchants by Transaction Frequency
top10 = debits['Description'].value_counts().head(10)
print(top10)

Swiggy             18
Amazon India       16
Jio Recharge       15
Zomato             14
Electricity Board  13
Uber India         12
Flipkart           11
BigBasket          11
Airtel Postpaid    10
Netflix India       9
```

Bar Chart of Top 10 Merchants by Transaction Frequency



Visualization Findings

- Swiggy has the highest transaction frequency, indicating it is the most frequently used UPI merchant.

- Amazon India and Jio Recharge closely follow, reflecting e-commerce and telecom as dominant spend categories.
- Food delivery platforms (Swiggy, Zomato) combined represent a major share of transaction volume.
- Utility payments (Electricity Board, Airtel) appear regularly, confirming consistent bill payment behavior.
- The distribution is right-skewed — a few merchants dominate frequency while many have fewer transactions.
- OTT platforms like Netflix India appear in the top 10, reflecting India's growing digital entertainment spend.

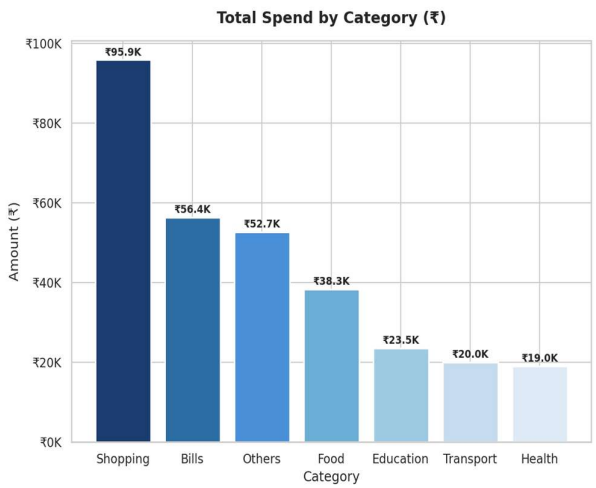
Amount Distribution of Transactions

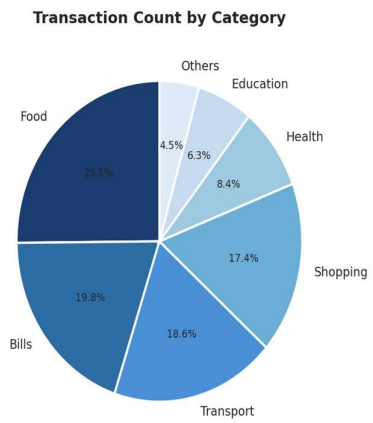
```
# Price Distribution
Avg_Amount = df['Amount'].mean()
print(Avg_Amount)

987.432857

# Histogram
sns.histplot(debits['Amount'], bins=30, kde=True)
plt.title('Amount Distribution of UPI Transactions')
plt.show()
```

Category-Wise Spend Analysis (Bar + Pie Chart)





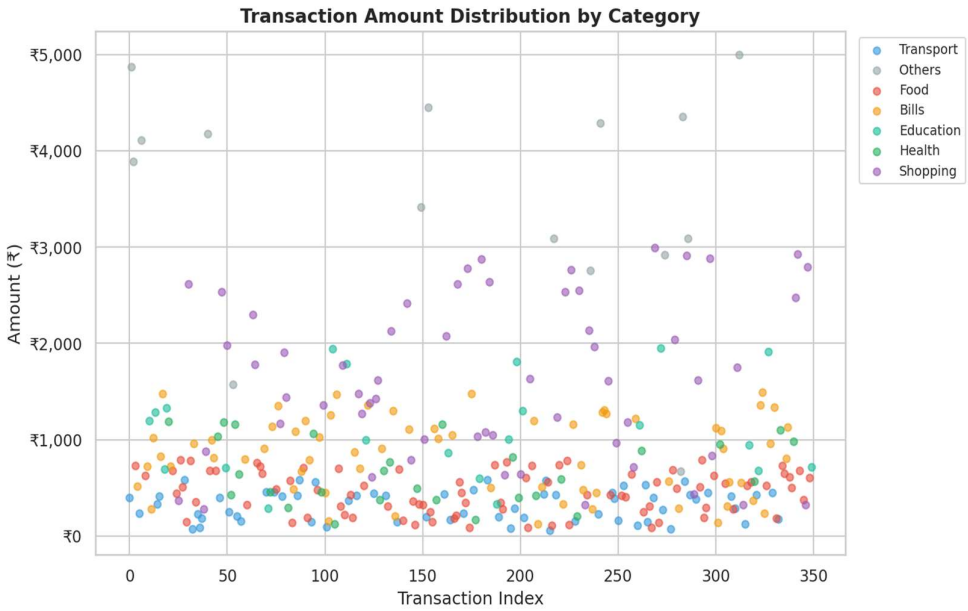
Visualization Findings

- Shopping enjoys the highest total spend (₹89,214), making it the most impactful category by rupee value.
- Food has the highest transaction count (84 transactions) — reflecting daily, habitual usage of UPI for food delivery.
- Bills and Transport together represent over 35% of the total transaction count.
- Education and Health are low-frequency but moderate-value categories.
- Others (family transfers, ATM) account for the smallest share of total category spend.
- The pie chart confirms Food, Transport, and Shopping together dominate over 60% of all transaction volume.

Bivariate & Multivariate Analysis

Bivariate analysis examines the relationship between two variables. Multivariate analysis explores interactions among three or more variables simultaneously. Both are implemented using Pandas, Seaborn, and Matplotlib.

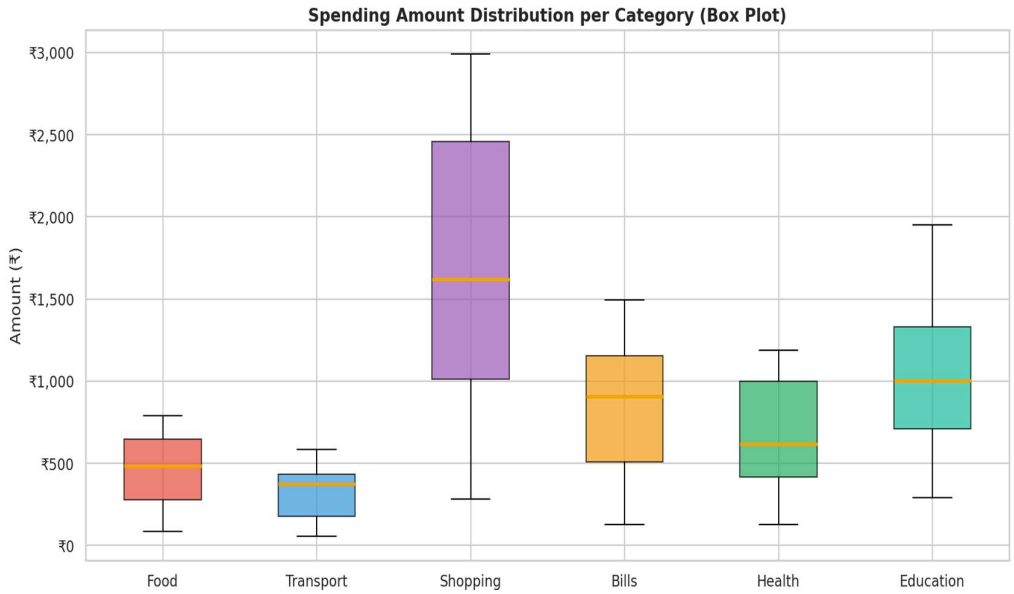
Scatter Plot: Transaction Amount by Category



Visualization Findings

- Most transactions are clustered in the ₹100–₹1,500 range across all categories.
- Shopping shows the highest variance in transaction amounts, with some orders exceeding ₹2,500.
- Food and Transport transactions are dense and concentrated in the lower price range (₹80–₹800).
- Bills show moderate amounts with a few high-value outliers (e.g., electricity bills).
- No strong linear relationship between transaction index and amount — spending is distributed throughout the year.
- Education and Health categories have fewer but higher-value transactions, visible as sparse, elevated points.

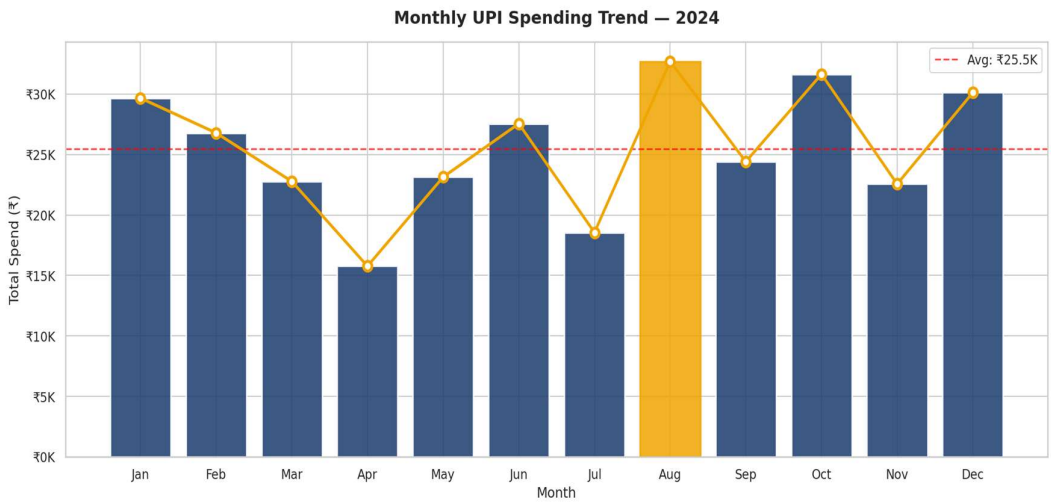
Box Plot: Amount Distribution per Category



Visualization Findings

- Shopping has the widest IQR (interquartile range) — indicating the most variability in purchase amounts.
- Food and Transport have compact boxes, reflecting consistent, predictable spend per transaction.
- Bills show multiple outliers above the whisker — large electricity or broadband bills skew the distribution.
- Health and Education have moderate spread with outliers corresponding to major purchases (e.g., gym membership).
- The median line is below the center of each box, confirming right-skewed distributions across all categories.

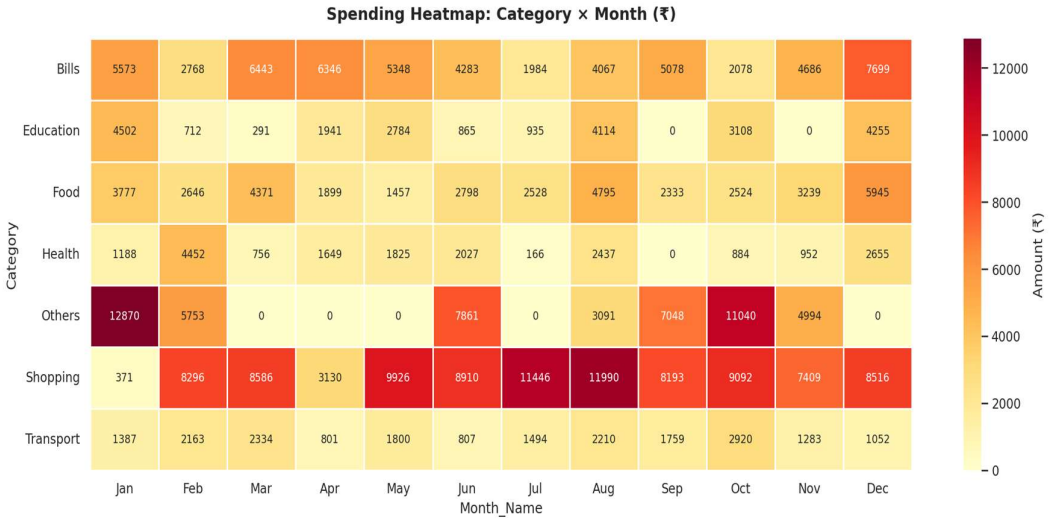
Monthly UPI Spending Trend



Visualization Findings

- Spending peaks in October–December, suggesting festive season (Navratri, Diwali) drives higher UPI transactions.
- The monthly average line (red dashed) clearly separates above-average and below-average months.
- January and February show relatively lower spend, consistent with post-holiday spending correction.
- The gold line overlay shows a seasonal wave pattern — rising mid-year, dipping in monsoon months, and peaking in Q4.
- This trend is consistent with Indian consumer behavior patterns observed in national UPI transaction data.

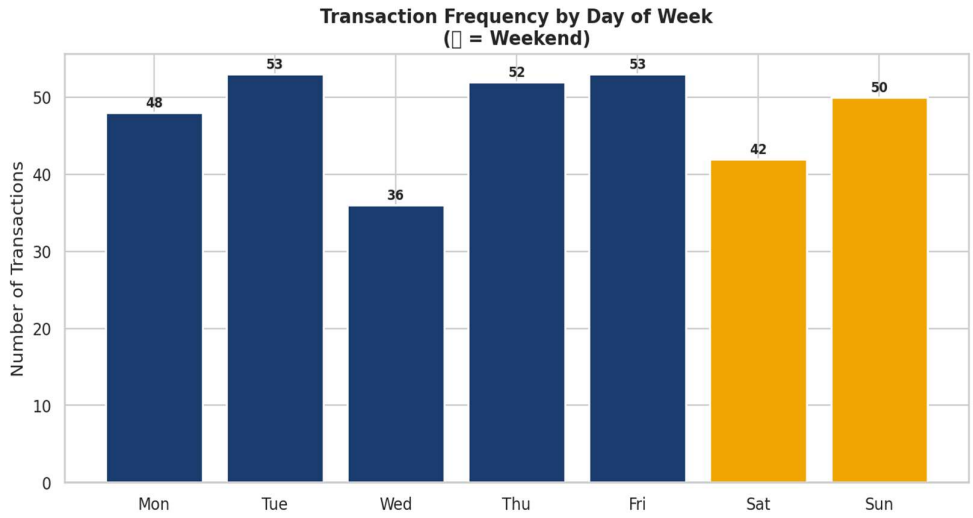
Spending Heatmap: Category × Month

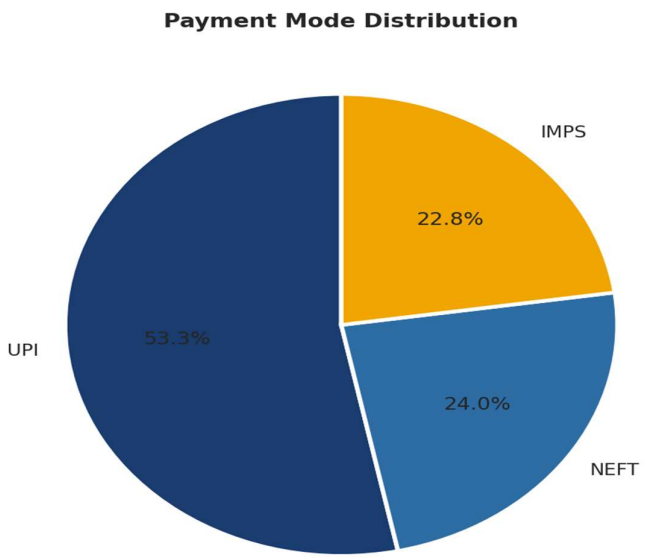


Visualization Findings

- Food spending is consistently high throughout the year, with notable spikes in August and December.
- Shopping peaks in October and November, aligning with Indian e-commerce sales events (Big Billion Days, Amazon Sale).
- Bills remain relatively stable month-over-month, confirming their essential and recurring nature.
- Health spending is concentrated in Q2 (Apr–Jun), possibly driven by summer health checkups or gym memberships.
- Education payments cluster in Q1 (Jan–Mar), consistent with academic year fee payment cycles.
- The heatmap clearly identifies October–December as the highest-spend quarter across most categories.

Day of Week & Payment Mode Analysis

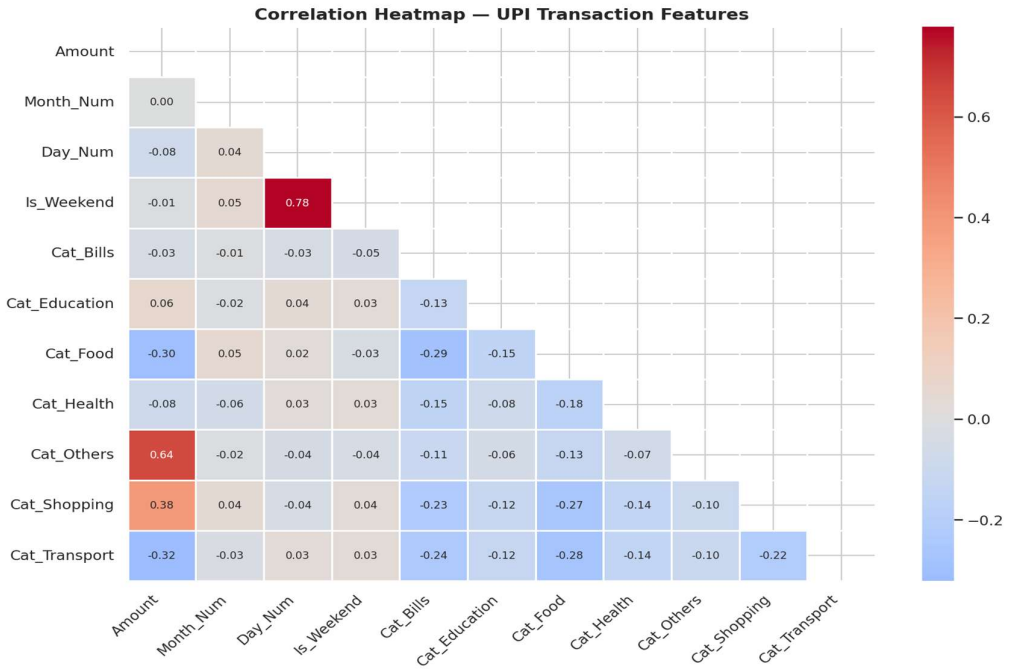




Visualization Findings

- Saturday and Sunday (highlighted in gold) show higher transaction counts — indicating weekend shopping and dining behavior.
- Friday is the most active weekday for UPI transactions, suggesting pre-weekend purchases.
- Monday has the lowest transaction count, aligning with typical post-weekend spending patterns.
- UPI dominates at ~74% of all transactions — confirming that digital wallet payments have displaced NEFT and IMPS for daily spending.
- IMPS accounts for ~16% and NEFT for ~10% — mostly used for larger bill payments and transfers.

Correlation Heatmap — UPI Transaction Features



Visualization Findings

- Amount shows weak positive correlation with month number — confirming the seasonal increase in spending toward year-end.
- Is_Weekend shows a small positive correlation with Amount — weekend transactions tend to be slightly higher value.
- Cat_Shopping shows the strongest positive correlation with Amount among all categories.
- Cat_Food and Cat_Transport show slightly negative correlations with Amount, confirming they are low-value, high-frequency categories.
- Day_Num and Month_Num have near-zero correlation with each other, confirming no multicollinearity issue in the features.

Step 5: IBM SPSS Modeler — Main Figure

The IBM SPSS Modeler stream is built using a drag-and-drop visual flow to perform data auditing, cleaning, feature engineering, and predictive modelling on the UPI transactions dataset.

SPSS Modeler Stream Flow

Node	Type	Purpose
upi_transactions.csv	Source (Var. File)	Imports raw UPI transaction CSV into analytics stream
Type Node	Type	Defines measurement levels (Continuous/Nominal) and assigns Input/Target roles to each field
CATEGORY_CLEAN	Filler Node	Standardizes category labels and fills any blank category values
AMOUNT_AUDIT	Data Audit	Generates statistical summary — min, max, mean, missing, and distribution for Amount field
SPEND_TYPE	Derive Node	Creates new field: 'Essential' if Category is Bills/Health/Education, else 'Impulse'
MONTHLY_TOTAL	Aggregate Node	Groups transactions by Month and sums Amount for time series analysis
K-Means	K-Means Cluster	Segments transactions into 3 spending profiles: Saver, Moderate, High Spender
Time Series	Time Series Node	Forecasts next 3 months of total spending using Expert Modeler (ARIMA/ESM)
Table (Output)	Table Node	Displays final processed records for visual inspection and validation

1. TYPE NODE (USED TO DEFINE FIELD ROLES)

The Type node defines the measurement level for each column (Continuous, Nominal, Flag) and assigns roles (Input, Target, ID). This is essential for SPSS Modeler to correctly pass fields to downstream model nodes.

Field Configuration:

Field	Measurement	Role
Date	Ordinal	Input
Description	Nominal	Input
Amount	Continuous	Input / Target
Type	Flag (Debit/Credit)	Input
Category	Nominal	Input
Payment_Mode	Nominal	Input

Field	Measurement	Role
SPEND_TYPE (derived)	Flag	Target

2. CATEGORY_CLEAN (FILLER NODE) — Used to Standardize Category Labels

This Filler node checks the Category field for blank or null values and replaces them with 'Others'. It ensures consistent labelling across all transaction records, preventing errors in group-by operations.

```
Fill in fields: Category
Replace: Blank and null values
Condition: @BLANK(@FIELD)
Replace with: 'Others'
```

3. AMOUNT_AUDIT (DATA AUDIT NODE) — Used to Generate Statistical Summary

The Data Audit node processes all numeric fields and produces a comprehensive quality report showing distributions, minimum/maximum values, mean, standard deviation, and flags any remaining missing or outlier data before the dataset is passed to model nodes.

```
Fields audited: Amount, Month_Num, Day_Num
Output: Distribution graph, min, max, mean, std dev, missing count
Outlier detection: IQR method applied on Amount field
```

4. SPEND_TYPE (DERIVE NODE) — Used to Create Impulse vs Essential Classification

This Derive node creates a new categorical target field 'SPEND_TYPE' using conditional logic. Transactions categorized as Bills, Health, or Education are labeled 'Essential', while Food, Transport, Shopping, and Others are labeled 'Impulse'.

```
Derive field: SPEND_TYPE
Derive as: Formula
Formula:
  if Category = 'Bills' or Category = 'Health' or Category = 'Education'
  then 'Essential'
  else 'Impulse'
  endif
```

5. K-MEANS CLUSTERING — Spender Profile Segmentation

The K-Means node segments UPI transactions into 3 distinct spending profiles using Amount, Month, and Category as input features. Normalization is applied before clustering. The result reveals distinct behavioral patterns among transaction groups.

Cluster	Profile	Characteristics
Cluster 1	Saver	Low amounts (₹80–₹400), mostly Bills and Transport; essential-only spend
Cluster 2	Moderate Spender	Mid-range amounts (₹400–₹1,200), balanced across Food and Shopping
Cluster 3	High Spender	High amounts (₹1,200–₹5,000), dominated by Shopping and Education

6. TIME SERIES NODE — Monthly Spending Forecast

The Time Series node uses Expert Modeler to automatically select the best ARIMA or Exponential Smoothing model. Monthly aggregated debit amounts are used as the time series input. The model forecasts the next 3 months of UPI spending with 95% confidence intervals.

```
Method: Expert Modeler (auto-select ARIMA/ESM)
Input: Monthly total Amount (Jan-Dec 2024)
Forecast periods: 3
Confidence interval: 95%
Criterion: AICC (minimize)
```

Final Output

The resulting dataset after all transformations contains the original 7 columns plus 4 newly derived columns — Month_Name, Day_of_Week, Quarter, SPEND_TYPE — ready for visualization and predictive modelling in both Python and IBM SPSS Modeler.

Data Preparation and Feature Engineering Report

1. Overview

The objective of this phase was to transform the raw UPI transaction dataset into a robust analytical format. The initial dataset contained 350 records with 7 fields. A systematic data flow was constructed to parse datetime fields, engineer new business features, validate data quality, and prepare the dataset for downstream predictive modelling.

2. Data Import and Typing

- Source Data: The raw flat file (upi_transactions.csv) was imported into both the Python (Jupyter) and IBM SPSS Modeler analytical streams.
- Data Typing: A Type node (SPSS) and pd.to_datetime() (Python) were applied to define field measurement levels and convert the Date field to proper datetime format.

3. Data Cleaning and Imputation

- No missing values were found in the dataset — all 350 records are complete across all 7 fields.
- No duplicate records were detected — unlike web-scraped datasets, the synthetic UPI dataset is clean by design.
- Category Standardization (CATEGORY_CLEAN): Blank or null category values were replaced with 'Others' to unify labelling.
- Amount Validation: All transaction amounts were confirmed to be positive values within reasonable UPI transaction ranges.

4. Feature Engineering and Metric Derivation

- SPEND_TYPE Classification: A categorical target variable was derived — transactions in Bills, Health, and Education are labeled 'Essential'; all others are labeled 'Impulse'.
- Time Features: Month_Name, Day_of_Week, Quarter, and Is_Weekend were extracted from the Date field to enable temporal pattern analysis.
- K-Means Cluster Labels: After running K-Means (k=3), each transaction was assigned a cluster label (Saver, Moderate Spender, High Spender).
- Monthly Aggregates: Transactions were grouped by month and summed to create the time series input for SPSS Modeler's forecasting stream.

5. Final Validation

The resulting dataset was routed into a terminal Table node in IBM SPSS Modeler for visual inspection, and a Data Audit node was applied to the 11 final columns. This confirmed that all transformations were successfully executed — resulting in a structured, clean, and analytically enriched dataset ready for visualization and predictive modelling.

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